

Autonomous Driving Agents for Safe Road Junctions Through Deep Reinforcement Learning

Jie Yun[†], Zehong Cao^{*}

^{*}University of South Australia, Adelaide, SA 5000, Australia

[†]University of Tasmania, Hobart, TAS 7000, Australia

Abstract—Autonomous driving agents have garnered significant public interest, particularly concerning their safety within transportation systems. Deep reinforcement learning (DRL) has emerged as a promising approach for developing these agents, especially in tasks such as obstacle avoidance. However, agents often operate with partial observations in complex traffic scenarios like road junctions, requiring DRL algorithms to rely on long-term rewards to train effective models. This reliance can introduce risks, particularly when agents explore new actions, potentially leading to collisions and penalties. Although safety considerations are typically integrated into reward functions, they are not always the primary metric for evaluating the effectiveness of DRL algorithms in autonomous driving agents. This study addresses this gap by assessing the comprehensive safety performance of three baseline DRL models, DQN, A2C, and PPO, based on collision, success, and freezing rates. We also propose a self-awareness DRL model incorporating a multi-attention mechanism for autonomous driving agents. Based on experiments in intersection and roundabout scenarios, the proposed model demonstrates enhanced safety performance compared to the best baseline model, DQN.¹

Index Terms—Autonomous Driving Agents, Deep Reinforcement Learning, Safe Road Junctions

I. INTRODUCTION

AUTONOMOUS driving technologies have garnered significant public interest due to their potential to bring substantial benefits such as reduced vehicle emissions and decreased traffic congestion [1], [2]. Despite these potential advantages, current self-driving agents remain at relatively early stages of development and must address various challenges, including efficient control, liability and safety [3], [4]. Traditional control theory and supervised learning have been central to developing path planning and control agents in autonomous driving [5], [6]. These methods, however, often rely on predefined models and extensive datasets, which can limit their adaptability in dynamic and unpredictable environments. In contrast, reinforcement learning (RL) has recently emerged as a powerful alternative. It allows vehicles to learn optimal control policies through direct interaction with their environment, eliminating the need for extensive prior knowledge and large training datasets [7]. Furthermore, deep reinforcement learning (DRL), an extension of RL, has achieved remarkable success in various domains, including autonomous driving. Notably, the deep Q-network (DQN) [8] introduced a novel approach by combining Q-learning with deep neural networks,

enabling the model to learn from raw sensory inputs and outperform human experts in complex video games. This success demonstrated the potential of DRL in environments where decision-making relies on processing high-dimensional data. Other DRL algorithms, such as advantage actor-critic (A2C) [9], optimise the training process by combining the benefits of value-based and policy-based methods to ensure efficiency in calculating value propositions. Additionally, proximal policy optimisation (PPO) [10], known for its robust convergence properties and ease of implementation, has been widely adopted in many RL-based agents, including traffic management and autonomous driving [11], [12]. These DRL-based approaches have significantly advanced state of the art in autonomous driving [13], but a notable limitation of these methods is their predominant focus on maximising cumulative rewards. This focus often comes at the expense of safety, a critical factor for the real-world deployment of autonomous driving agents. The challenge lies in balancing the pursuit of reward optimisation with the imperative to ensure the safety of autonomous driving agents, particularly in complex and dynamic traffic environments.

II. RELATED WORK

To achieve higher rewards, a vehicle must continuously learn from its experiences, often at the cost of collisions, to develop autonomous driving capabilities. Recent research in DRL-based driving has focused on long-term cumulative rewards or average rewards as crucial performance metrics [14]. However, safety measures have been largely overlooked to evaluate the autonomous performance of DRL models. While a recent study [15] addressed vehicle safety, it was limited to a straight four-lane highway and did not explore more complex road traffic junction scenarios. To investigate the potential of self-awareness in DRL, we examined the self-attention mechanism, first introduced by [16], to address challenges in translating complex sentences in natural language processing tasks. This mechanism allows the model to retain long input sequences by generating a dynamic context vector with importance weights. Building on this, [17] developed the transformer model, a multi-head self-attention module that performed exceptionally in translation tasks. Additionally, [18] adopted a variant of the transformer model to handle the varying sizes of input data from surrounding vehicles in autonomous driving. This approach enhanced the model's ability to interpret interactions between traffic participants

¹Zehong Cao is corresponding author. This work is partially supported by ARC DECRA Fellowship DE220100265.

and improved the representation of features in a function approximation setting. However, current research has not yet developed an attention mechanism that directly addresses the safety concerns of autonomous vehicles, particularly in challenging road traffic junction scenarios.

In this work, we aim to (1) evaluate safety concerns using four metrics—collision rate, success rate, freezing rate, and reward—across three state-of-the-art DRL models (DQN, A2C, and PPO) in complex driving environments, such as intersections and roundabouts; and (2) propose a self-awareness module for DRL models to enhance safety performance in these challenging scenarios. The outcomes of this research will contribute to developing safer DRL approaches for autonomous driving agents.

III. METHODOLOGY

As shown in **Figure 1**, we propose a self-awareness safety DRL, attention-DQN, in a simulated road traffic junction environment including intersection and roundabout scenarios. The ego vehicle is an autonomous driving agent who observes the state s_t of the environment and decides an action a_t ; for instance, the car accelerates in the forward direction. The environment then gives a reward r_t as feedback and enters a new state s_{t+1} . This autonomous driving agent contains two deep networks: the Q and target networks. The Q network includes all updates during training, while the target network is used to retrieve Q values. The target network parameters are updated only with the Q network parameters every 512 steps to reduce the correlation between Q-value updates and thus make the Q-network output's distribution stationary [19]. DQN also uses the concept of experience replay to form a sufficiently stable input dataset for training, i.e., all experiences in the form of $e_t = (s_t, a_t, r_t, s_{t+1})$ are stored in the replay memory and sampled uniformly in a random manner. We expect the self-awareness safety mechanism driven by attention modules to allow the ego vehicle to filter relevant information and prioritise vehicles that may pose potential conflicts with its planned route. This enhanced situational awareness enables the ego vehicle to make more informed decisions, thereby reducing the risk of collisions and improving overall safety. We present this in more detail as follows.

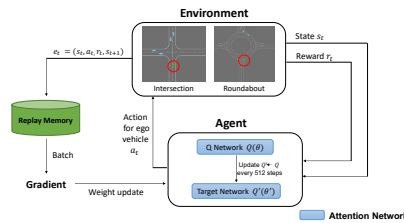


Fig. 1. Our proposed self-awareness safety DRL (attention-DQN).

DQN [8] is a foundational DRL model for autonomous driving agents inspired by [20], where the goal of the agent is to interact with the environment by selecting actions that maximise cumulative future rewards. The DQN approximates the optimal action-value function as follows:

$$Q^*(s, a) = \mathbb{E}_{s' \sim \epsilon} \left[r + \gamma \max_{a'} Q^*(s', a') | s, a \right] \quad (1)$$

where $Q^*(s, a)$ is the maximum sum of rewards discounted by γ at each time step, achievable by following any strategy in a state s and taking action a . This function follows the Bellman equation: if the optimal value $Q^*(s', a')$ of the state s' at the next time step is known for all possible actions, then the optimal strategy is to select the action a' that maximises the expected value of $r + \gamma Q^*(s', a')$.

The Q-network updates only the target values periodically to reduce correlations with the target, which is trained by minimising the following loss function L at iteration i :

$$L_i(\theta_i) = \mathbb{E}_{(s,a,r,s') \sim U(D)} \left[(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i))^2 \right] \quad (2)$$

where θ_i denotes the parameters of the Q-network at iteration i . θ_i^- are the parameters of the target network at iteration i , which are held fixed between individual updates and are updated only with the Q-network parameters θ_i every specified number of steps.

Furthermore, the architecture of an attention network is illustrated in **Figure 2**. For an attention network, the observations containing vehicle features (i.e., location and velocity) are split into the ego vehicle's inputs and those of other surrounding vehicles. The input observation information is normalised and then passed to a multilayer perceptron (MLP) network; then, the outputs from both MLPs are passed to a multi-head (i.e., two heads) attention block. Specifically, the multi-head attention block contains a single query Q from the ego vehicle, the keys of all the vehicles K and the values of all the vehicles V . Linear transformations are applied to Q , K , and V , as well as the scale dot-product attention that is calculated from the transformed data. Then, the attention data from multiple heads are concatenated and linearly transformed before being passed to the next layer. Afterwards, the attention data from multiple heads are combined and added to the MLP output of the ego vehicle and then fed to a LayerNorm layer. Finally, the regularised data from the normalisation layer are passed to the output MLP to produce the action values.

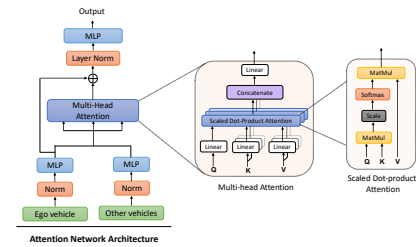


Fig. 2. The architecture of a self-attention network.

In addition, we introduce **Figure 3** to present further how an attention head works. In terms of the single-query attention architecture shown in **Figure 3-A**, first, the ego vehicle emits a default single query $Q = [q_0]$ to select a subset of vehicles according to the context. The query is computed with a linear

projection L_q . Second, the query is compared to a set of keys $K = [k_0, \dots, k_N]$ that contain descriptive features k_i for each vehicle, and the keys are computed with a shared linear projection L_k . Third, the dot product of the query and keys QK^T is computed to assess the similarities between them. The result is scaled by $(1/\sqrt{d_k})$, where d_k is the dimension of the keys, and a softmax function is applied to obtain the weights on the values $V = [v_0, \dots, v_N]$. The values are also computed with a shared linear projection L_v . Notably, the keys K and values V are concatenated from all vehicles, whereas the query Q is produced by only the ego vehicle. The output attention matrix is formalised as:

$$\text{output} = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

This extends to a multi-query attention architecture, marked as "MultiQ-attention-DQN", as demonstrated in **Figure 3-B**, which was adapted from the single-query attention architecture to observe the performance variations and the main differences were highlighted in orange. Here, we include queries from all the vehicles instead of using the default single query from the ego vehicle.

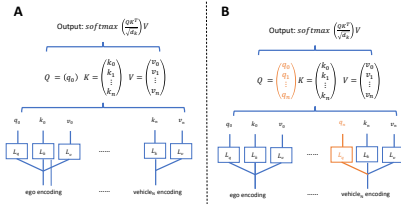


Fig. 3. The architecture of an attention head: (A) single-query attention head; (B) multi-query attention head.

IV. EXPERIMENTS

A. Driving Environment

In this study, we used two representative driving scenarios, intersection and roundabout, as shown in **Figure 4**, to investigate the safety of DRL for road traffic junction environments based on the highway simulation system [21]. We considered intersection (**Figure 4-A**) and roundabout (**Figure 4-B**) scenarios characterised as relatively challenging driving interflow environments. The scenarios include one ego vehicle trying to cross traffic and arrive at a destination and several surrounding vehicles spawned randomly. The ego vehicle needs to make decisions such as turning left, changing lanes, accelerating and decelerating while avoiding colliding with other vehicles.

For the intersection scenario, the scene comprises two roads crossing perpendicularly and stretching in four directions (north, south, east, and west). The roads are populated with the ego vehicle and several surrounding vehicles. The positions, velocities and destinations of the other vehicles are randomly initialised. The ego vehicle drives from 60 metres south to the intersection, and the task of the ego vehicle is to cross the intersection and turn left (west). The goal is achieved (i.e., destination reached) if the ego vehicle can turn left at the

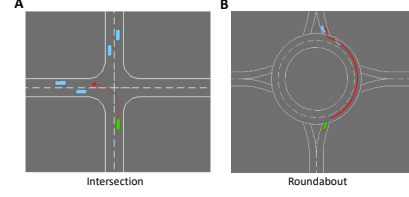


Fig. 4. Two representative driving scenarios in the road traffic junctions: (A) intersection; (B) roundabout. The ego vehicle with green colour is driving as guided by the red arrow to reach the goal.

intersection and drive 25 metres or more from the turning point in 13 seconds. **Figure 4-A** is a showcase of the intersection scenario, where the goal of the ego vehicle shown in green is to turn left, as indicated by the red arrow. The vehicle observations of the intersection scenario follow the kinematic bicycle model, which implements simplistic behaviour and considers the same lane interactions to prevent potential lateral collisions [21]. The ego vehicle selects one from a finite set of actions $A = \{\text{slower}, \text{no-operation}, \text{faster}\}$, where the vehicle can choose to slow down, maintain a constant speed or accelerate. The ego vehicle receives a reward of 1 if it drives at the maximum speed or if it arrives at the destination and receives a reward of -5 if a collision occurs. This reward design encourages the ego vehicle to cross the intersection and arrive at the destination as soon as possible while simultaneously avoiding collisions.

The roundabout scenario is composed of a two-lane roundabout with four entrances/exits (north, south, east, and west), and the roads are occupied by the ego vehicle and several surrounding vehicles. The positions, velocities and destinations of the other vehicles are initialised randomly. The ego vehicle drives from 125 metres south to the roundabout, and the task of the ego vehicle is to cross the roundabout and take the second exit (north). The goal is accomplished if the ego vehicle successfully takes the second exit at the roundabout and drives 10 metres or more from the exit point in 13 seconds. As presented in **Figure 4-B** for a showcase of the roundabout scenario, the goal of the ego vehicle is to exit the roundabout following the red arrow. The ego vehicle observes and learns from a $V * F$ array, where V is the number of nearby vehicles and F is the size of the features observed. In our setting, the ego vehicle observes a feature set of size 7, $\{p, x, y, v_x, v_y, \cos_h, \sin_h\}$, where p represents whether a vehicle is in the row (whether a vehicle was observed), x and y describe the location of the vehicle, v_x and v_y denote the x and y axes of velocity, and $\cos_h$ and $\sin_h$ describe the heading direction of the vehicle. The actions of the ego vehicle select from a finite set $A = \{\text{lane_left}, \text{lane_right}, \text{idle}, \text{faster}, \text{slower}\}$, implying that the vehicle can choose to change lanes to the left/right, maintain the same speed and lane, and accelerate or decelerate. The ego vehicle receives a reward of 0.5 if it drives at the maximum speed and receives a reward of -1 if a collision occurs. Every time the vehicle changes lanes, it will be awarded -0.05. We also applied a success reward of 1 if the ego vehicle arrives at the destination. This reward design

encourages the vehicle to cross the roundabout and arrive at the destination as soon as possible while simultaneously avoiding collisions and unnecessary lane changing.

B. Evaluation Metrics

Our study defined four evaluation metrics that strongly consider safety concerns (collision, success, and freezing) and the standard total reward to assess three baseline DRL models (DQN, PPO, and A2C) and our proposed self-awareness safety DRL: attention-DQN.

- *Collision rate.* It calculates the percentage of episodes in which a collision occurs with the ego vehicle.
- *Success rate.* It counts the percentage of episodes in which the ego vehicle successfully reaches the destination within the allowed duration. It also considers the functionality of the autonomous vehicle, which evaluates the ability of a DRL model to learn behaviour to complete the desired task. There is a timeout setting representing the maximum duration to cross the intersection, where exceeding this duration is considered unacceptably slow.
- *Freezing rate.* It determines the percentage of episodes in which the ego vehicle freezes or moves extremely slowly in the driving environment. The ego vehicle can become a serious safety hazard to traffic, such as the case of causing serial rear-end collisions, and the freezing rate is calculated based on the residual value from the success rate and collision rate as follows:

$$Rate_{freezing} = 100\% - Rate_{collision} - Rate_{success} \quad (4)$$

- *Total reward.* It is a typical metric commonly used to evaluate ego vehicle premium performance over several episodes. Our experiment collected the total reward that the ego vehicle receives after taking a set of actions in each episode.

C. Experimental Procedure

This study involved two experiments. In Experiment 1, we trained the ego vehicle using three baseline DRL models, DQN, A2C, and PPO and evaluated their safety performance across the intersection and roundabout scenarios using four key metrics. In Experiment 2, we compared the performance of the proposed single-query and multi-query attention-DQN models with the baseline DQN model in the same driving scenarios. Both experiments assessed safety performance during the training and testing phases.

1) *Experiment 1:* In the training phase, the ego vehicle was trained using the baseline DRL models (DQN, A2C, PPO) across 60,000 episodes with 5 runs. Collision and success rates converged during this period. In the testing phase (validation), each model was evaluated over 100 episodes for 100 trials, recording collisions and successes to calculate collision, success, and freezing rates.

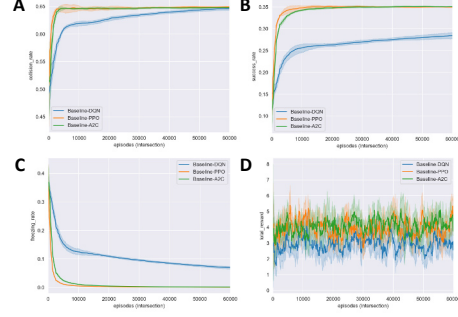


Fig. 5. Intersection scenario: the training performance of the ego vehicle with respect to four metrics: (A) collision rate, (B) success rate, (C) freezing rate, and (D) total reward using the baseline DRL models (DQN, A2C, and PPO).

2) *Experiment 2:* We trained the ego vehicle using single-query and multi-query attention-DQN models, also over 60,000 episodes with 5 runs, using the same hyperparameters as the baseline DQN for fair comparison. The testing phase (validation) followed the same procedure as in Experiment 1, where collision, success, and freezing rates were recorded for performance evaluation.

V. RESULTS

A. Results in Experiment 1

- *Intersection scenario.* In **Figure 5**, we presented the training evolution performance of the collision rate (**Figure 5-A**), success rate (**Figure 5-B**), freezing rate (**Figure 5-C**) and total reward (**Figure 5-D**) of the ego vehicle in the intersection scenario, trained by three baseline DRL models (DQN, PPO, and A2C) over 60,000 episodes and repeated 5 times, and the displayed values show the averages with the 95% confidence intervals. The collision rate of the ego vehicles trained by the three baseline DRL models exceeded 60% at the end of the training process, particularly in DQN, which had the lowest collision rate during the training process. Moreover, the success rates of PPO and A2C reached approximately the same value of 35%, and the success rate of DQN reached approximately 28%, as reflected by the total reward curves. In addition, the freezing rates of the ego vehicles trained by PPO and A2C reached approximately 0, starting from 15,000 episodes, whereas the freezing rate of DQN decreased to approximately 8%, achieving the best performance among all baseline DRL models.
- *Roundabout scenario.* In **Figure 6**, we demonstrated the training evolution of the performance of the collision rate (**Figure 6-A**), success rate (**Figure 6-B**), freezing rate (**Figure 6-C**) and total reward (**Figure 6-D**) of the ego vehicle during training in the roundabout scenario, trained by three baseline DRL models (DQN, PPO, and A2C) over 60,000 episodes and repeated 5 times, and the displayed values show the averages with the 95% confidence intervals. The collision rate of the ego vehicles trained by all baseline DRL models was below 25% at the end of the training, with A2C achieving the lowest value of approximately 5%.

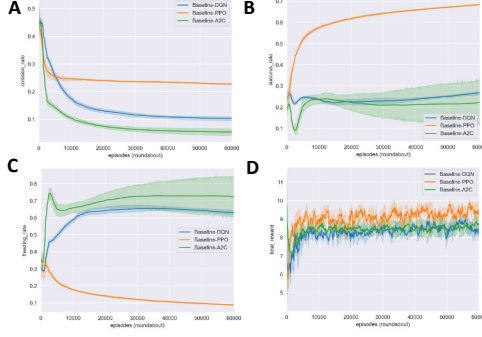


Fig. 6. Roundabout scenario: the training performance of the ego vehicle with respect to four metrics: (A) collision rate, (B) success rate, (C) freezing rate, and (D) total reward using the baseline DRL models (DQN, A2C, and PPO).

In contrast, the freezing rates of the ego vehicles trained by A2C and DQN both exceeded 60%, with that of A2C being approximately 73% and that of DQN being approximately 63%. In addition, PPO achieved the lowest freezing rate and the highest success rate among the baseline DRL models.

- **Validation.** **Table I** shows the collision rates, success rates, and freezing rates of the ego vehicles tested in the intersection and roundabout scenarios using three baseline DRL models over 100 episodes and averaged across 100 runs (10,000 testing episodes in total for each model). Regarding the intersection scenario, the collision rates of the ego vehicles trained by all three baseline models exceeded 50%, with PPO being the highest at 64.45% and DQN being the lowest at 56.88%. In terms of success rate, DQN also had the lowest value of 29.23%. This was caused by DQN being the only case that experienced a freezing robot problem, with a freezing rate of 13.89%. The PPO and A2C models did not have freezing vehicles during the testing phase. Moving to the testing phase of the roundabout scenario, A2C demonstrated the lowest collision rate of 7.2%, followed by DQN at 14.7% and PPO at 20.47%. DQN had the lowest success rate and the highest freezing rate, whereas PPO had the highest success rate and lowest freezing rate, with A2C being the second in both metrics.

TABLE I
THE TESTING PERFORMANCE OF THE THREE BASELINE DRL MODELS IN THE INTERSECTION AND ROUNDABOUT SCENARIOS.

Scenarios	Metrics mean (std) %	DQN	A2C	PPO
Intersection	Collision Rate	56.88 (5.39)	65.51 (5.08)	64.45 (4.93)
	Success Rate	29.23 (4.99)	34.49 (5.08)	35.54 (4.93)
	Freezing Rate	13.89 (3.81)	0 (0)	0 (0)
Roundabout	Collision Rate	14.7 (7.24)	7.2 (3.03)	20.47 (4.78)
	Success Rate	33.99 (5.66)	46.35 (7.70)	74.42 (4.91)
	Freezing Rate	51.31 (10.33)	46.45 (8.58)	5.11 (5.39)

B. Results in Experiment 2

- **Intersection scenario.** In **Figure 7**, we presented the training evolution of the collision rate (**Figure 7-A**), success rate (**Figure 7-B**), freezing rate (**Figure 7-C**) and total reward

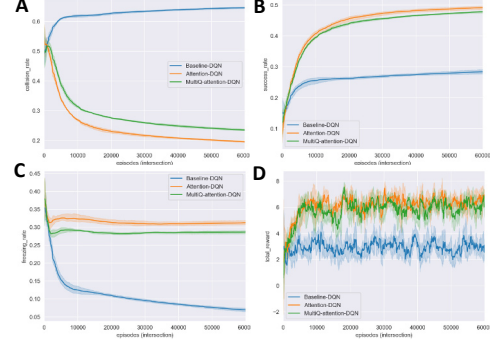


Fig. 7. Intersection scenario: the training performance of the ego vehicle with respect to four metrics: (A) collision rate, (B) success rate, (C) freezing rate, and (D) total reward using DQN, single-query and multi-query attention-DQN.

(**Figure 7-D**) of the ego vehicle in the intersection scenario, learned by DQN, default (single-query) and multiQ (multi-query) attention-DQN over 60,000 episodes. The figure shows the mean values averaged from 5 runs with 95% confidence intervals. The ego vehicle trained by the default attention-DQN (orange colour) achieved the lowest collision rate and the highest success rate, which was demonstrated by a significant drop in the collision rate from approximately 67% for DQN to approximately 20% for attention-DQN, as well as by a considerable increase from 28% for DQN to 49% for the attention-DQN. The changes in collision and success rates were also reflected in the total reward. Additionally, the multiQ-attention-DQN (green colour) exhibited similar performance but not as strong as that of the default attention-DQN.

- **Roundabout scenario.** **Figure 8** demonstrates the training evolution of the collision rate (**Figure 8-A**), success rate (**Figure 8-B**), freezing rate (**Figure 8-C**) and total reward (**Figure 8-D**) of the ego vehicle in the roundabout scenario learned by DQN, default (single-query) and multiQ (multi-query) attention-DQN over 60,000 episodes. The figure shows the mean values averaged from 5 runs with 95% confidence intervals. The ego vehicle trained by attention-DQN was quickest to avoid collisions during the beginning of the training period, as indicated by the curve's steepness between episodes 1 and 10,000, and then reached a low collision rate of 12% at episode 60,000. For the success rate, the ego vehicle trained by the attention-DQN achieved the highest performance of approximately 74%, which is a significant improvement compared to the 27% achieved by DQN. In addition, the freezing rate dropped from 63% for DQN to approximately 15% for attention-DQN.
- **Validation.** In **Table II**, we summarised the collision rate, success rate, and freezing rate of the ego vehicle tested by trained DRL models (DQN, default attention-DQN and MultiQ-attention-DQN) over 100 episodes and averaged across 100 runs (10,000 testing episodes in total for each DRL model) in the intersection and roundabout scenarios. In the intersection scenario, the testing performance of the ego vehicle trained by attention-DQN showed the lowest

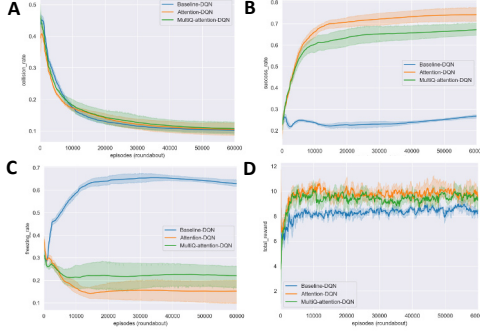


Fig. 8. Roundabout scenario: the training performance of the ego vehicle with respect to four metrics: (A) collision rate, (B) success rate, (C) freezing rate, and (D) total reward using DQN, single-query and multi-query attention-DQN.

collision rate of 14.57% and the highest success rate of 52.58%. For the roundabout scenario, similarly, attention-DQN achieved the lowest collision rate of 12.80% and presented a significant increase in the success rate, from 33.99% for DQN to 83.92% for attention-DQN. The freezing rate of the attention-DQN unveiled the lowest value of 3.28%, which was reduced significantly compared to that of DQN (51.31%). Furthermore, the performance of multiQ-attention-DQN ranked second, slightly worse than attention-DQN, but very close.

TABLE II
THE TESTING PERFORMANCE OF DQN, SINGLE-QUERY AND MULTI-QUERY ATTENTION-DQN IN INTERSECTION AND ROUNDABOUT SCENARIOS.

Scenarios	Metrics mean (std) %	DQN	Attention-DQN	
			Default	MultiQ
Intersection	Collision Rate	56.88 (5.39)	14.57 (3.76)	18.54 (6.40)
	Success Rate	29.23 (4.99)	52.58 (6.35)	49.18 (6.74)
	Freezing Rate	13.89 (3.81)	32.85 (6.03)	32.28 (8.88)
Roundabout	Collision Rate	14.70 (7.24)	12.80 (5.30)	12.93 (4.73)
	Success Rate	33.99 (5.66)	83.92 (4.30)	80.90 (8.06)
	Freezing Rate	51.31 (10.33)	3.28 (3.17)	6.17 (9.93)

VI. CONCLUSION

In this study, we evaluated the safety performance of autonomous driving agents in a complex road traffic junction environment, including intersection and roundabout scenarios. We considered comprehensive safety evaluation metrics, including collision rate, success rate, freezing rate, and total reward, to assess the current DRL models (DQN, A2C, and PPO) and our proposed attention-DQN module with self-awareness capability. Our results based on two experiments showed the unsafe cases of the current DRL models, while our proposed DRL can significantly reduce collision rate and freezing rate by introducing the attention mechanism, validating the effectiveness of our proposed attention-DQN module in enhancing the safety concerns of the ego vehicle.

Our findings can potentially contribute to and benefit safe DRL in future transportation applications.

REFERENCES

- [1] M. Li, Z. Cao, and Z. Li, "Augmented mixed vehicular platoon control with dense communication reinforcement learning for traffic oscillation alleviation," *IEEE Internet of Things Journal*, 2024.
- [2] M. Li, Z. Li, and Z. Cao, "Enhancing car-following performance in traffic oscillations using expert demonstration reinforcement learning," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [3] B. Chen and Z. Cao, "Hlg: Bridging human heuristic knowledge and deep reinforcement learning for optimal agent performance," in *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*, 2024, pp. 2189–2191.
- [4] B. Chen, Z. Cao, and Q. Bai, "Satf: A scalable attentive transfer framework for efficient multiagent reinforcement learning," *IEEE Transactions on Neural Networks and Learning Systems*, 2024.
- [5] D. Stavens, G. Hoffmann, and S. Thrun, "Online speed adaptation using supervised learning for high-speed, off-road autonomous driving," in *IJCAI*, 2007, pp. 2218–2224.
- [6] R. Hussain and S. Zeadally, "Autonomous cars: Research results, issues, and future challenges," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 2, pp. 1275–1313, 2018.
- [7] C. Wu, A. Kreidieh, K. Parvate, E. Vinitsky, and A. M. Bayen, "Flow: Architecture and benchmarking for reinforcement learning in traffic control," *arXiv preprint arXiv:1710.05465*, vol. 10, 2017.
- [8] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [9] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, "Asynchronous methods for deep reinforcement learning," in *International conference on machine learning*, PMLR, 2016, pp. 1928–1937.
- [10] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [11] M. Li, Z. Cao, and Z. Li, "A reinforcement learning-based vehicle platoon control strategy for reducing energy consumption in traffic oscillations," *IEEE Transactions on Neural Networks and Learning Systems*, 2021.
- [12] Z. Cao and C.-T. Lin, "Reinforcement learning from hierarchical critics," *IEEE Transactions on Neural Networks and Learning Systems*, 2021.
- [13] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. A. Al Sallab, S. Yogamani, and P. Pérez, "Deep reinforcement learning for autonomous driving: A survey," *IEEE Transactions on Intelligent Transportation Systems*, 2021.
- [14] M. Zhou, J. Luo, J. Villella, Y. Yang, D. Rusu, J. Miao, W. Zhang, M. Alban, I. Fadarar, Z. Chen et al., "Smarts: Scalable multi-agent reinforcement learning training school for autonomous driving," *arXiv preprint arXiv:2010.09776*, 2020.
- [15] J. Rong and N. Luan, "Safe reinforcement learning with policy-guided planning for autonomous driving," in *2020 IEEE International Conference on Mechatronics and Automation (ICMA)*. IEEE, 2020, pp. 320–326.
- [16] J. Chorowski, D. Bahdanau, D. Serdyuk, K. Cho, and Y. Bengio, "Attention-based models for speech recognition," in *Advances in Neural Information Processing Systems*, 2015, pp. 577–585.
- [17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [18] E. Leurent and J. Mercat, "Social attention for autonomous decision-making in dense traffic," *arXiv preprint arXiv:1911.12250*, 2019.
- [19] G. Li, R. Gomez, K. Nakamura, and B. He, "Human-centered reinforcement learning: A survey," *IEEE Transactions on Human-Machine Systems*, vol. 49, no. 4, pp. 337–349, 2019.
- [20] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller, "Playing atari with deep reinforcement learning," *arXiv preprint arXiv:1312.5602*, 2013.
- [21] E. Leurent, "An environment for autonomous driving decision-making," *GitHub repository*, 2018.